
Class Diagrams and Domain Models

Submission Instructions:

- This is an individual assignment.
- Submit your own original work. **Plagiarism will not be tolerated at all**, either done from the internet or from any fellow.
- Any submission that does not adhere to these instructions will not be evaluated.

Scenario 1:

An international airline manages flights, dynamic rebookings, and VIP loyalty programs. Flights can be merged (e.g., two flights combine mid-journey due to weather rerouting). Loyalty points can be transferred to family members but only after validating compliance rules.

→ Q1:

Design a **domain model** that:

- Captures flight merging and rebooking scenarios.
- Models loyalty points transfer with validation constraints.
- Shows multiplicity and conditional associations (e.g., transfer conditions).

Scenario 2:

A healthcare platform connects patients, wearable devices, and doctors. Devices can push real-time data to doctors, but devices occasionally **update firmware**, changing their reporting structure. Doctors may "subscribe" only to specific types of updated data streams (e.g., heart

rate but not glucose).

→ Q2:

Create a **class diagram** that models:

- Devices with dynamic firmware versions.
- Patient-device associations.



● Subscription filtering by doctors for specific data types.

Scenario 3:

A freelancing platform allows freelancers to join multiple professional guilds. Each guild grants certifications based on skill tests, but **certifications must be renewed** based on annual activity points. Some projects require freelancers with **multi-guild** certifications.

→ Q3:

Model a **domain model** that:

- Connects freelancers, projects, certifications, and guilds.
- Captures certification renewal based on freelancer activity.
- Handles projects requiring cross-guild validation.

Scenario 4:

In a smart city, autonomous vehicles (buses, emergency drones) can **switch control** between AI and human operators based on emergency levels. A legal audit system logs when and why control was transferred.

→ Q4:

Design a **class diagram** showing:

- Vehicles with dual control states (AI vs Human).
- Emergency escalation triggers.
- Audit log records of all control transfer events.

Scenario 5:

A museum exhibits both physical artifacts and VR experiences. Each VR experience can branch into alternate history versions based on visitor interactions. Visitors can influence the exhibit and leave a "footprint" (record) of how they changed the narrative.

→ Q5:

Develop a **domain model** representing:

- Artifacts, VR twins, and branching historical narratives.
- Visitor interactions as trigger
- Footprints capturing interaction history.

